

Hospital Management Dashboard

Project Report

An interactive Power BI dashboard for monitoring patient demographics, doctor availability, billing, and payment trends

Tool Used	: Microsoft Power BI
Domain	: Healthcare Analytics
Dashboard Pages	: 2 (Main Dashboard + Detailed Metrics/Overview)
Focus Areas	: Patient Demographics, Doctor Load, Billing, Payments

September 12, 2026

Contents

1	Introduction	2
2	Scope of Analysis	2
3	Methodology & Approach	2
4	Dashboard Features	3
5	Key Insights	3
6	Conclusion	3

1 Introduction

Hospitals generate continuous streams of clinical and financial data — patient admissions, doctor assignments, billing, and payments — but this data is only useful if it can be reviewed quickly and acted upon. This project builds an interactive **Power BI dashboard** that consolidates hospital operations and financial performance into a single, easy-to-navigate view for hospital management and stakeholders.

Objective

The objective of this project is to design an interactive Hospital Management Dashboard using Power BI that provides a comprehensive overview of hospital operations and financial performance. The dashboard helps hospital management and stakeholders monitor patient demographics, doctor availability, billing status, and payment trends — supporting data-driven decision-making, improving operational efficiency, and enhancing financial control.

Problem Statement

"To build a Power BI dashboard that analyzes hospital patient data, doctor specialization, and billing performance to support operational and financial decision-making."

2 Scope of Analysis

The dashboard was designed to answer the following operational and financial questions for hospital administrators:

- What is the total patient volume, and what does the patient age profile look like?
- How many doctors are available, and how is patient load distributed across specializations?
- What is the total billed amount, and how much of it has been collected vs. is still pending?
- Which payment modes (Cash, Card, Online, Insurance, etc.) are most commonly used by patients?
- Where are the biggest opportunities to improve collections and resource allocation?

3 Methodology & Approach

- **Data Preparation:** Performed ETL and data wrangling to clean and integrate structured hospital datasets (patient records, doctor assignments, billing data) into a model suitable for Power BI reporting.
- **Data Modeling:** Built relationships between patient, doctor, and billing tables, and created DAX measures to calculate key metrics efficiently and optimize dashboard performance.
- **Dashboard Design:** Designed a **2-page interactive Power BI dashboard** — a main dashboard view for at-a-glance KPIs, and a detailed metrics/overview page for deeper drill-down analysis.
- **Visualization:** Used KPI cards, bar/column charts, and breakdown visuals to represent patient demographics, doctor specialization load, and payment mode distribution in an intuitive, decision-ready format.

Technical Stack

- **Tool:** Microsoft Power BI
- **Techniques:** ETL, Data Wrangling, Data Modeling, DAX Measures, Dashboard Design

4 Dashboard Features

Metric / View	Purpose
Total Patients & Average Age	Provides insight into patient demographics and case mix
Total Doctors	Helps management assess resource availability and specialization coverage
Total Bill Amount	Tracks overall hospital revenue generated from billing
Paid Amount vs. Pending Amount	Highlights collected revenue vs. outstanding dues for financial control
Doctor Specialization vs. Patient Load	Identifies high-demand specialties and supports better doctor allocation
Payment Mode Analysis	Breaks down payments by Cash, Card, Online, Insurance, etc. to understand patient payment behavior

5 Key Insights

- **Patient Demographics:** Analyzing average patient age and total patient volume together gives hospital management a clear view of the case mix they are serving, which informs staffing and resource planning.
- **Doctor Utilization:** Comparing doctor specialization against patient load surfaces which specialties are under high demand and which have spare capacity — enabling more balanced doctor allocation.
- **Financial Health:** Separating the total bill amount into paid vs. pending gives administrators immediate visibility into cash flow and outstanding dues, rather than a single opaque revenue figure.
- **Collections Strategy:** Payment mode analysis (Cash, Card, Online, Insurance) reveals which channels patients prefer and where billing/collection processes could be streamlined – e.g., encouraging digital payment modes if online/card show faster settlement than cash.
- **Operational Transparency:** Consolidating clinical and financial metrics into one interactive dashboard replaces scattered manual reporting with a single source of truth for hospital administration.

6 Conclusion

This project demonstrates the effective use of Power BI for healthcare analytics — combining data modeling, DAX-driven calculations, and visualization to turn raw hospital data into an actionable, two-page interactive dashboard.

By consolidating patient demographics, doctor availability, billing amounts, and payment mode analysis into a single view, the dashboard enables hospital administrators to monitor financial health and reduce pending payments, optimize doctor utilization across specializations, gain insight into patient trends, and improve transparency in operational planning.

“Developed an interactive hospital management dashboard analyzing patient demographics, doctor availability, billing amounts, payment modes, and pending dues, providing actionable insights for hospital administration.”

Future Scope:

- Add trend-over-time views (monthly/quarterly) for patient volume and revenue to track seasonal patterns.

- Introduce automated alerts for pending payments crossing a defined threshold.
- Extend the model to include bed occupancy and department-wise resource utilization.